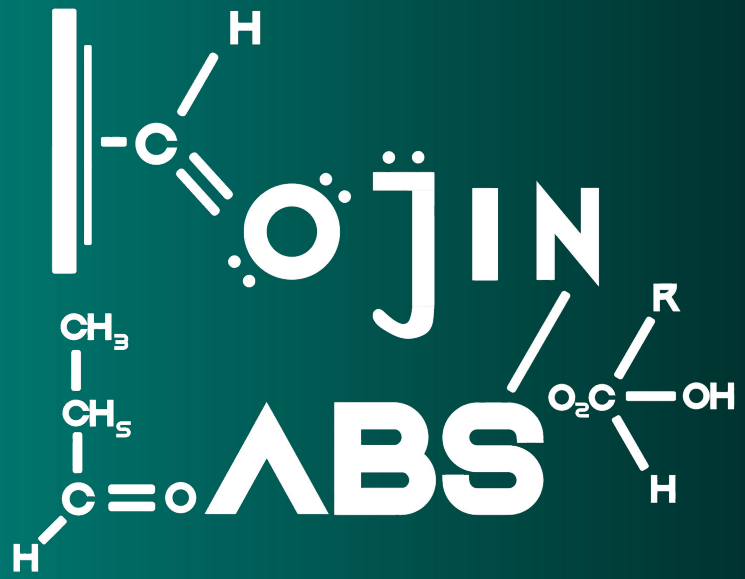


Kojin Labs

In pursuit of cleaner air for a healthier society.

Our Team Dynamic

Strengths

- Communication
- Supportive/ psychologically safe space
- Deep passion

Growth Areas

- Focus on feasibility
- Specify/ work out the details
- Accountability

Problem Identified: The indoor air problem

- Formaldehyde can enter indoor air from the natural chemical decomposition of building materials, furniture, adhesives, as well as the process of combustion.
- Particle filtration alone does not address gaseous pollutants.

We set out to answer this question: Where could targeted treatment meaningfully reduce exposure?

Our initial scope

Primary target

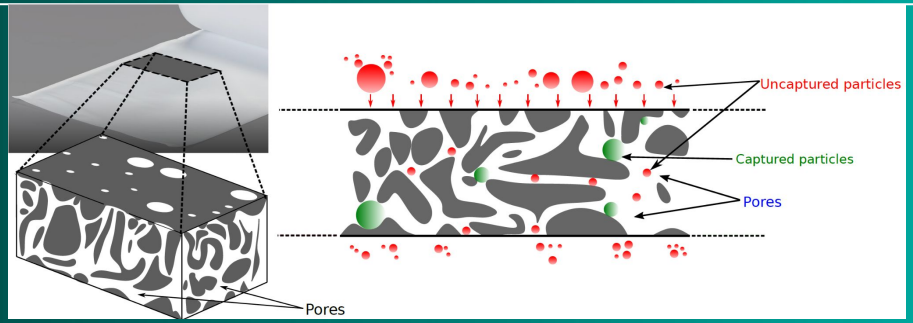
Formaldehyde in occupied indoor spaces

Proposed format

A replaceable gas-treatment layer for an air system

Fall focus: literature review, stakeholder interviews, and measurement planning
Later: supervised bench tests, then a pilot program if research is fruitful

Why the Status Quo needs Innovation:



Cross sectional area of common activated carbon filters (particulate only)

- Particle filters leave a gap.
- Existing treatments have practical limits.
- Real-world conditions matter.

Our opportunity: Investigate a compact treatment layer that reduces exposure while preserving airflow and keeping maintenance affordable.

95%

Sampled homes exceeded a chronic formaldehyde benchmark[1]

6.7×

Price delta between gas and particle filtration setups[2]

Meet Elena

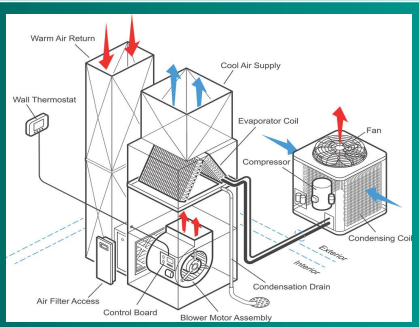
- 38 years old
- Furniture restoration specialist
- Poorly ventilated workshops
- Experienced:
  - airborne formaldehyde
  - toxic aldehyde vapors

→

Diagnosis: nasopharyngeal carcinoma



Stakeholders



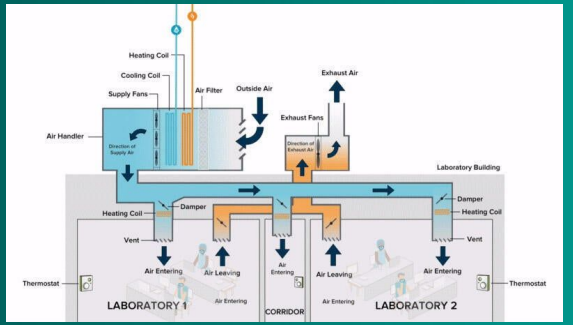
Occupants and housing communities → Exposure concerns, affordability, and control over indoor conditions

Facilities and HVAC teams → Airflow limits, installation, maintenance, and purchasing

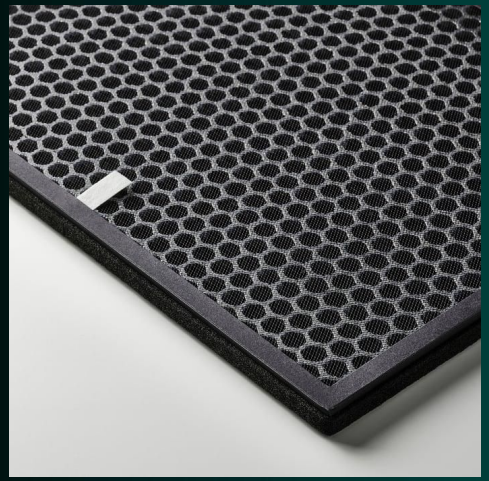
Health and measurement specialists → Sampling methods, exposure benchmarks, and safety

Materials researchers and filter suppliers → Chemistry, stability, manufacturing, and existing alternatives



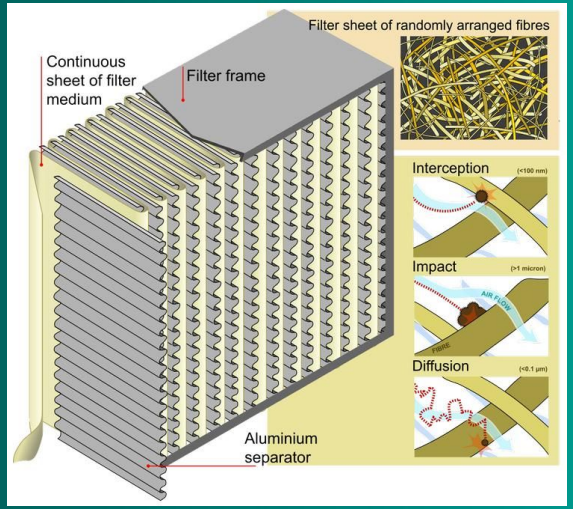


Current Solutions



Activated Carbon air purifiers

- carbon traps certain gas molecules
- limited capacity
- expensive replacement cost
- less effective under high heat



High-Efficiency Particulate Air (HEPA) filters

- dense network of fibers that capture particles
- does not reliably capture formaldehyde gas/ gaseous VOCs
- require separate gas-phase treatment

THANK YOU!

Want to learn more?



Sources

1. Mullen, Nasim, et al. “Impact of Natural Gas Appliances on Pollutant Levels in California Homes.” *Earnest Orlando Lawrence Berkley National Labratory*, 1 Dec. 2012, doi:10.2172/1163521.
2. https://www.iqair.com/products/air-purifiers/gc-multigas?Title=Default+Title and https://cowaymega.com/products/mighty-ap-1512hh